Research Article

Coaching DNA: Measuring Strategic Identity and Knowledge Transmission in the NFL

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Abstract: We develop a framework to measure coaching "DNA" in the NFL, the strategic tendencies that distinguish how coaches run teams. Using XGBoost models trained on more than 600,000 plays from 2006–2024 with a leakage-free, cross-fit pipeline, we quantify four dimensions of coaching identity: offensive aggression (fourth down, pass-run balance, deep passing, two-point), shotgun formation preference, tempo, and defensive aggression. Each gene is the residual between actual decisions and situation-adjusted predictions, separating tendency from forced circumstance. We first show the genes are stable coach traits, not team artifacts: coach identity explains about five times the team's variance, and the genes travel across team changes (composite offensive aggression $r = 0.61$, 95% CI $[0.33, 0.78]$; shotgun 0.55, $[0.29, 0.75]$). Under coach-clustered inference, composite offensive aggression ($r = 0.30$, $[0.16, 0.45]$) and shotgun preference ($r = 0.23$, $[0.04, 0.41]$) predict career-level, context-adjusted Coach Wins Above Replacement (Coach WAR), while tempo does not. The offensive aggression-WAR association has weakened in recent seasons, but within coaches higher offensive aggression still adds wins ($\beta = 1.40$ WAR games per standard deviation, $[0.76, 1.96]$). Neither mentor performance ($r = 0.12$, $[-0.01, 0.26]$) nor situational aggression ($r = 0.08$, $[-0.05, 0.22]$) transmits to protégés, but shotgun preference transmits through offensive coaching trees ($r = 0.40$, $[0.21, 0.54]$) and persists from coordinator to head coach ($r = 0.67$, $[0.48, 0.80]$). Coaching trees transmit systems (shotgun and defensive aggression, $r = 0.57$–0.67, $p \leq 0.02$) far more than risk tolerance or winning, which barely transmit (mentor WAR $r = 0.12$; situational aggression $r = 0.08$).

Keywords: NFL coaching, coaching DNA, coaching trees, strategic identity, competitive advantage

1 Introduction

Fourth-down analytics consistently show that aggressive coaches win more games (Romer, 2006; Yam and Lopez, 2019). Teams that "go for it" on fourth down in marginal situations would gain approximately 0.4 additional wins per season on average (Yam and Lopez, 2019). Expected points models demonstrate that punting from midfield often surrenders value (Burke, 2009). These findings raise a deeper question in competitive strategy: if aggressive decisions are individually optimal, does their advantage persist once every team adopts them?

NFL coaching is among the highest-stakes decision-making environments in professional sports. Head coaches command salaries that average several million dollars and reach $20 million for the highest-paid (Sportico, 2024), and their individual in-game choices are dissected in granular detail by analysts and fans. Yet how a coach is characterized, as aggressive, conservative, run-heavy, or innovative, still rests largely on narrative and reputation rather than measurement. We lack an objective, decision-level vocabulary for a coach's strategic tendencies.

These tendencies surface in dozens of observable choices every game: whether to go for it on fourth down, whether to run from shotgun or under center, how fast to snap the ball, how many defenders to send after the quarterback. Each choice reflects a coach's strategic identity, their "DNA." But the tendencies are hard to measure objectively, hard to separate from forced circumstance, and hard to classify as stable traits versus situational adaptations. Building that vocabulary is the goal of this project.

1.1 Measuring Coaching Philosophy

We propose a general framework for quantifying coaching tendencies across multiple strategic dimensions. For each decision type, we train a machine learning model on situational features (game state, field position, score, time, environment) to predict what the "average" coach would do in identical circumstances. Depending on the decision, this model is a classifier (for discrete choices such as going for it on fourth down) or a regressor (for continuous behaviors such as seconds between snaps); the gene is defined identically in both cases. A coaching gene is then measured as the systematic deviation between a coach's actual decisions and these situation-adjusted predictions:

$$\text{Gene}_c = \text{Actual Rate}_c - \text{Predicted Rate}_c \tag{1}$$

This residual approach isolates genuine strategic tendency from forced circumstance. A coach who goes for it on fourth down frequently only because they are always trailing is not truly aggressive. Only the *excess* tendency beyond what the situation demands is attributed to coaching philosophy. We apply this framework to four dimensions of coaching DNA:

- **Offensive Aggression**: a composite of four sub-components, namely fourth-down decisions, pass-run balance, deep passing tendency, and two-point conversion attempts. This is the dimension with the strongest theoretical prior for predicting performance, since analytics research finds aggressive fourth-down decisions are frequently expected-value optimal (Burke, 2009; Romer, 2006).
- **Shotgun formation preference**: a single gene capturing whether coaches run more or less shotgun than game context predicts, reflecting offensive system design independent of situational necessity.
- **Tempo**: a composite of no-huddle usage and pace (seconds between plays, with faster-than-expected scored positively), measuring how fast an offense operates.
- **Defensive aggression**: a composite of box stacking, pass rush intensity, and man-versus-zone coverage tendency, measured at the team level from 2016 onward (man/zone from 2018).

Three of the four dimensions are composites of their sub-components and shotgun preference is a single gene; together they profile coaching DNA across both sides of the ball.

1.2 The Coaching Tree Framework

When NFL teams hire head coaches, pedigree frequently influences decisions. Coordinators from successful organizations receive premium consideration, with the implicit assumption that working under successful coaches transfers strategic expertise and winning approaches. The coaching tree hypothesis makes three testable predictions: (1) successful mentors produce successful protégés, (2) strategic tendencies transmit through apprenticeship, and (3) coaching DNA persists through promotion from coordinator to head coach. We directly test all three.

1.3 WAR as the Measure of Coaching Success

To test whether coaching genes relate to success, we need a measure of coaching quality that separates a coach's contribution from inherited circumstances, since win-loss records confound coaching ability with roster talent, strength of schedule, and luck. We adopt Coach Wins Above Replacement (Coach WAR), a context-adjusted estimate of the wins a coach adds relative to a replacement-level alternative, developed and validated on held-out coaches in companion work (Author, 2026). Using Coach WAR as the outcome lets us ask whether a coach's strategic DNA predicts winning once roster quality and circumstance are accounted for.

1.4 This Study

This study defines these genes and tests three claims about them:
1. **Real**: the genes are stable traits that belong to coaches rather than to teams.
2. **Predictive**: a coach's strategic DNA predicts coaching success, measured by Coach WAR, beyond inherited roster quality.
3. **Heritable**: the genes transmit from mentors and coordinators to the head coaches they produce.

We find that the genes are real and that several predict winning, but that coaching trees pass on systems far more reliably than success or risk tolerance.

2 Literature Review

The study of coaching decision-making in the NFL has focused primarily on conservative bias. Romer (2006) demonstrated that NFL coaches punt too frequently on fourth down, forgoing positive-expected-value conversion attempts, a robust and widely cited finding. Yam and Lopez (2019) extended this work with causal estimates, finding that teams adopting more aggressive fourth-down strategies would gain approximately 0.4 wins per season. Horowitz and Shim (2014) confirmed the persistence of the fourth-down puzzle using different statistical methods, while Burke (2009) generalized the finding through expected points models showing that conservative play-calling surrenders value across a range of game situations. Collectively, this literature establishes that aggressive coaching decisions are often optimal in isolation, but does not address whether the advantage persists as adoption spreads.

A parallel literature has sought to quantify team quality more broadly. Schatz (2003) developed DVOA (Defense-adjusted Value Over Average), one of the first publicly available metrics for evaluating team performance adjusted for opponent quality. These metrics, however, are fundamentally team-level *outcome* measures that combine the on-field performance of players, coordinators, and coaching staffs. They capture how well a team performed, not the strategic choices behind that performance, and so cannot separate a head coach's philosophy from the talent executing it. On coaching careers specifically, Goff and Wisley (2006) identified a managerial life cycle in the NFL, in which coaches tend to perform best in their early years and decline over time, suggesting that coaching effectiveness is not static and may interact with organizational context. What remains missing is a framework that measures coaching *decision tendencies* rather than outcomes, namely what coaches choose to do beyond what the situation demands, decomposed into dimensions that can be tracked across time, relationships, and career transitions.

The competitive dynamics of strategic adoption have been studied extensively outside sports. Rogers (2007) formalized the S-curve pattern of innovation diffusion, in which early adopters gain outsized advantages that erode as the innovation becomes mainstream. Because the analytics literature establishes that aggressive tactics are expected-value positive, this framework predicts that aggressive tactics should follow a similar trajectory: they provide a competitive edge while rare, an edge that erodes as analytics-driven decision-making becomes universal. However, this prediction has not been empirically tested in the coaching context.

Several gaps emerge from this literature. No studies have examined how the offensive aggression-performance relationship evolves temporally as analytics disseminate and defenses adapt. Offensive Aggression persistence within coaches, distinguishing stable traits from situational tactics, has not been systematically analyzed. Coaching tree effects have not been tested using context-adjusted performance metrics that control for inherited roster quality. Whether strategic philosophies genuinely transmit from mentors to protégés, and which dimensions show transmission, remains unexamined.

3 Methods

3.1 Data

Our analysis integrates three complementary datasets spanning nearly two decades of NFL football. The primary dataset consists of play-by-play records for every offensive snap from the 2006 through 2024 seasons, obtained through the `nfl_data_py` package. This yields more than 600,000 analyzable plays with complete situational information (down, distance, field position, score differential, time remaining, and environmental conditions). The exact number of plays entering each model varies with the decision being modeled. The 2006 starting point is determined by data availability, particularly for air yards data required for our deep passing analysis.

The coaching data comes from career histories scraped from Pro Football Reference, giving complete position-by-position timelines for each coach. From these we build the coaching tree, the network of mentor-protégé relationships on which our inheritance analyses rest, matched on a year-aware canonical franchise key so that relocations are handled correctly. We scope the tree to the modern era (1970–2025): 325 coaches connected by 1,425 documented mentor-protégé relationships, where a coordinator or position coach is counted as a protégé of any head coach they worked under for at least one season. Coach backgrounds (offensive, defensive, or other) are assigned by pattern matching on the role titles a coach held before becoming a head coach. The gene measures described below, by contrast, require play-by-play data and so exist only for 2006–2024, a window in which 185 of these coaches were active; analyses that need a gene value are restricted accordingly.

Performance metrics come from a separate Wins Above Replacement (WAR) calculation that estimates each head coach's contribution to team success relative to a replacement-level alternative (Author, 2026). This dataset contains 1,637 coach-year observations spanning the modern era (1970–2024), which we merge with our gene measures on a canonicalized coach name and season, logging attrition. The merge yields 608 coach-year observations for offensive genes (offensive aggression, shotgun, tempo) and 282 for defensive aggression (2016–2024 only), covering 124 unique head coaches; the 33 unmatched gene coach-years (31 coaches) are all partial-season interim stints (none a full 16-game season). Coach WAR does not score these stints because they lack a replacement-level baseline, and in their gene values they do not differ systematically from the retained sample. We classify these coach-years by background as 310 offensive (51.2%) and 251 defensive (41.4%), with 45 (7.4%) having substantial experience on both sides of the ball; the offensive-versus-defensive comparisons that follow use the 561 cleanly classified observations.

3.2 Measuring Coaching Genes

Each gene measures systematic deviation from expected decision-making. Rather than using raw frequencies, which confound individual tendencies with situational factors, we train machine learning models to predict baseline behavior and measure how much a coach or team deviates from these predictions. Across the four gene dimensions we fit ten component models, one per observable decision, and form the genes from their residuals. Offensive Aggression, tempo, and defensive aggression are *composite* genes built from several component models, while shotgun preference is a single-model gene; Table 1 maps the ten models to the four genes.

We use gradient-boosted decision trees (XGBoost) for every component model because they are consistently the strongest performers on tabular, point-in-time prediction of this kind: with heterogeneous structured features rather than images or text, gradient-boosted trees outperform deep networks (Chen and

Guestrin, 2016; Grinsztajn et al., 2022). Each model is a classifier or a regressor depending on its target, but the residual definition of a gene (Equation 1) is identical in both cases. Every model is trained on situational features alone and excludes season, week, and coach identity, so the baseline reflects what the average coach would do in identical circumstances and our analyses of temporal evolution and individual differences are not confounded. Hyperparameters are tuned per model by randomized search (100 iterations), with 5-fold stratified cross-validation for classifiers and KFold for regressors.

Tab. 1: The four coaching genes and their ten component models

Gene	Component model	Type	Attribution	Features	Seasons
Offensive Aggression (composite)	Fourth-down decision	Classifier	Head coach	16	2006–2024
	Run vs. pass	Classifier	Head coach	16	2006–2024
	Pass target (deep)	Classifier	Head coach	17	2006–2024
	Two-point conversion	Classifier	Head coach	16	2006–2024
Shotgun (single)	Shotgun formation	Classifier	Head coach	16	2006–2024
Tempo (composite)	No-huddle	Classifier	Head coach	18	2006–2024
	Pace (seconds/play)	Regressor	Head coach	15	2006–2024
Defensive aggression (composite)	Box stacking	Regressor	Team-year	15	2016–2024
	Pass rush	Regressor	Team-year	17	2016–2024
	Man coverage	Classifier	Team-year	20	2018–2024

The Features column gives the number of stability-selected predictors each model retains from the shared candidate pool (Table 2). Offensive Aggression, tempo, and defensive aggression are composites of their component genes; shotgun is a single component.

Tab. 2: Shared candidate feature pool for the component models

Feature Category	Models
Game State (13 features) Score differential, quarter, seconds remaining, timeouts remaining (both teams), down, distance	All
Field Position (5 features) Yard line, goal-to-go, side of field, distance, down	All except Two-Point
Environmental (6 features) Home/away, division game, roof, surface, temp, wind	All
Drive Context (3 features) Drive play count, drive first downs, net yards	All
Vegas Lines (2 features) Spread, total	All
Pre-snap indicators (2 features) Shotgun, no-huddle (observed pre-snap offense)	Defensive models

Offensive models draw on up to 26 candidates (21 for two-point, which omits field position); defensive models add the two pre-snap indicators for 28. Each model's stability-selected subset is listed in Table 1.

From the candidate pool in Table 2, each model retains only a stability-selected subset of features (see Section 3.3), reducing the 21–28 candidates to 15–20 stable predictors per model and guarding against overfitting collinear context variables.

Model performance varies across the ten components and is summarized for all of them in Table 3. Under grouped resampling that holds out whole team-years (so no team's plays appear in both training and evaluation), the classifiers range from man coverage (AUC 0.68) up to fourth-down decisions (AUC 0.98), while the regressors explain a more modest share of variance. In every case, situational factors explain most of the decision-making variance, leaving a smaller residual that we attribute to coaching philosophy.

Tab. 3: Predictive Model Performance (Nadeau-Bengio corrected resampling)

Classifiers (AUC)

Gene	Model	AUC	95% CI	Seasons
Offensive Aggression	4th Down Decision	0.975	[0.973, 0.976]	2006–2024
	Run vs Pass	0.784	[0.782, 0.786]	2006–2024
	Pass Target	0.734	[0.731, 0.736]	2006–2024
	Two-Point Conversion	0.927	[0.914, 0.940]	2006–2024
Shotgun	Shotgun Formation	0.807	[0.802, 0.812]	2006–2024
Tempo	No-Huddle	0.766	[0.758, 0.775]	2006–2024
Def. Aggression	Man Coverage	0.680	[0.673, 0.687]	2018–2024

Regressors (RMSE and R^2)

Gene	Model	RMSE	95% CI	R^2 [95% CI]	Seasons
Tempo	Pace (seconds)	11.08	[11.04, 11.11]	0.428 [0.424, 0.432]	2006–2024
Def. Aggression	Box Stacking	0.811	[0.801, 0.820]	0.422 [0.411, 0.433]	2016–2024
	Pass Rush	0.811	[0.793, 0.829]	0.055 [0.051, 0.059]	2016–2024

Point estimates and 95% CIs are Nadeau-Bengio corrected resampled-t intervals (Nadeau and Bengio, 2003) over $J = 50$ grouped resamples (GroupShuffleSplit holding out whole team-years, so no team appears in both train and test). The corrected variance inflates the naive resample variance to account for the overlap among resampled training sets. RMSE units differ by target: seconds for pace, players for box stacking and pass rush. The low pass-rush R^2 reflects that pass-rusher counts are only weakly predictable from pre-snap context, leaving more of their variation to coaching tendency.

3.2.1 Offensive Aggression Gene

Offensive Aggression is a composite of four component models, each predicting a discrete in-game decision: going for it on fourth down, passing versus running, targeting beyond the first-down marker versus at or short of it (air yards exceeding the yards to go), and attempting a two-point conversion versus an extra point. For each coach-year, we calculate component-specific offensive aggression scores using Equation 1, where c indexes the four components. The composite offensive aggression score is a *reliability-weighted* average of the four standardized component scores. Each coach-year's component is weighted by its reliability, the share of its variance that is true between-coach signal rather than sampling noise, so that a component measured from few plays in a given coach-year contributes less. We estimate these reliabilities with a DerSimonian-Laird random-effects model (DerSimonian and Laird, 1986), detailed in Section 3.3. We treat offensive aggression as a *formative* construct. The four components are distinct facets that jointly constitute aggressive decision-making rather than indicators of a single latent trait (they are only weakly inter-correlated), so we report component-level results alongside the composite throughout.

3.2.2 Shotgun Formation Gene

We train an XGBoost classifier to predict whether each play will be run from shotgun formation, using the offensive candidate feature set (Table 2, stability-selected to 16 features). The shotgun gene for each coach-year is the difference between actual and predicted shotgun rates, reliability-shrunk toward the league mean (so low-play coach-years are not over-trusted) and z-scored across all coach-years for comparability; because head-coach seasons carry ample plays, this shrinkage is small in practice. This gene captures formation preference independent of game context and is attributed to head coaches, yielding 638 coach-year observations across 2006–2024.

3.2.3 Tempo Gene

The tempo gene combines two sub-components into a composite measure. First, a no-huddle classifier predicts whether each play will use a no-huddle formation, capturing deliberate no-huddle usage beyond what game context (such as trailing late) predicts. Second, a pace regressor predicts the number of seconds between consecutive plays using the same candidate set with no-huddle excluded (to avoid circularity). The pace gene is the negated residual, so positive values indicate faster-than-expected pace. It captures play clock management philosophy. First plays of each drive are excluded from the pace calculation as they have no meaningful inter-play interval.

The composite tempo gene is the reliability-weighted mean of the z-scored no-huddle and pace sub-components, attributed to head coaches. This composite captures overall pace philosophy: coaches who both run more no-huddle and snap the ball faster score highly, while coaches who are fast on one dimension but slow on the other receive intermediate scores. The composite yields 638 coach-year observations across 2006–2024.

3.2.4 Defensive Aggression Gene

Defensive aggression is a composite of three component models, each capturing an aggressive defensive tendency at the team level: box stacking (loading more defenders near the line than offensive formation and game context predict, a hallmark of run-stuffing fronts), pass rush intensity (sending more rushers on passing plays), and man coverage (playing man rather than zone, the more aggressive, talent-dependent option). We measure defensive aggression at the team level rather than attributing it to an individual coach, reflecting the shared influence of head coach, defensive coordinator, and available personnel. Box stacking and pass rush are available from 2016 and man coverage from 2018, when these tracking and coverage-classification fields enter `nfl_data_py`. The composite is the reliability-weighted mean of the z-scored components, using two components for 2016–2017 and three from 2018; because a mean of two unit-variance components has a different scale than a mean of three, we standardize within each component-count regime so that pre- and post-2018 team-years share a common scale (eliminating an artificial break at 2018). This yields 288 team-year observations across 2016–2024, with the head coach of each defending team recorded for inheritance analysis.

3.3 Modeling Pipeline and Feature Selection

Several methodological safeguards apply uniformly to all ten predictive models and are essential for the validity of the resulting genes.

Because a gene is the residual between observed behavior and a model's prediction, the gene is only meaningful if predictions at scoring time are generated in exactly the feature space the model was trained in. We therefore fit all preprocessing on the training split alone. The categorical encoders, the median imputer, and the truncated-SVD reconstruction step are estimated on training data, persisted, and then applied in transform-only mode when computing genes for the full play-by-play record. This eliminates a train/serve mismatch in which preprocessing re-estimated on all plays would place genes in a different feature space than the one the models learned. Two further leakage paths are closed elsewhere in this section. Coach identity, season, and week are excluded from every model as predictors, so a model cannot recover a coach's tendencies from their name or from the calendar; and every data split is made at the team-year group level, so no coach or team appears in both training and evaluation (detailed below).

We do not hand-curate the full feature set: a pre-specified core of game-state fields is always retained, and the remaining context variables are selected by stability selection (Meinshausen and Bühlmann, 2010). For each model we draw repeated subsamples of the data, subsampling at the team-year group level so that a team's or coach's plays are never split across the in- and out-of-sample sets, then refit the model and rank features by gain importance. For shortlist length K and subsample b, let $\hat{S}_K^{(b)}$ denote the K highest-importance features; the selection frequency of feature k is

$$\hat{\Pi}_k^{(K)} = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\left\{k \in \hat{S}_K^{(b)}\right\}, \tag{2}$$

over $B = 50$ subsamples, each a 50% sample of team-year groups drawn without replacement. We treat a feature as stable when its selection frequency reaches the threshold $\pi = 0.6$, and take the stable set as the union of such features over the shortlist lengths $K \in \{8, 12, 16\}$. This prunes the 21–28 candidate context variables (several of them collinear, such as the three time-remaining measures and the score-differential family) to a parsimonious 15–20 stable predictors per model, reducing variance from redundant inputs without materially changing predictive accuracy. We protect a pre-specified core of fine-grained game-state fields (the clocks, score margin, field position, down and distance, and timeouts) from pruning so that the situational baseline a gene is measured against is always fully controlled.

A gene compares a coach's behavior to a model's prediction, so if the model were trained on that same coach its prediction would be partly fit to the coach's own tendencies, shrinking the residual and attenuating every gene toward zero. We remove this by cross-fitting: predictions used to compute genes are out-of-fold, generated under group k-fold cross-validation that leaves out whole coaches (offensive genes) or whole teams (defensive genes), reusing the tuned hyperparameters. Each coach-year is therefore scored by a model that never saw that coach, so the residual reflects genuine deviation from a coach-blind baseline. Folds are formed by coach or team rather than by season; because every model excludes season and week and pools plays across all years, its baseline is a single all-era norm, so era-imbalanced folds do not introduce time-shift bias into the residuals. This matters most for the small-sample components, whose residuals would otherwise be attenuated toward zero.

Each composite gene (offensive aggression, tempo, defensive aggression) is a reliability-weighted mean of its z-scored sub-components; the single-component shotgun gene uses the analogous empirical-Bayes shrinkage. For a sub-component, a coach-year's residual θ_i has binomial sampling variance $v_i = \sum \hat{p}(1-\hat{p})/n_i^2$, and we estimate the between-coach-year variance with the DerSimonian-Laird random-effects estimator (DerSimonian and Laird, 1986),

$$\hat{\tau}^2 = \max\left(0,\ \frac{Q-(m-1)}{C}\right), \tag{3}$$

where $Q = \sum_i w_i(\theta_i - \bar{\theta})^2$ is Cochran's statistic, $w_i = 1/v_i$, $\bar{\theta} = \sum_i w_i\theta_i / \sum_i w_i$, $C = \sum_i w_i - \sum_i w_i^2 / \sum_i w_i$, and m is the number of coach-years. Each coach-year's component then enters the composite with weight equal to its reliability,

$$\mathrm{rel}_i = \frac{\hat{\tau}^2}{\hat{\tau}^2 + v_i}, \tag{4}$$

so a component measured from few plays contributes less. Because the genes are residuals from classifier probabilities, we verified that the classifiers are well-calibrated out-of-fold (expected calibration error ≤ 0.03 for all seven), so the residuals are not driven by systematic mis-calibration. Because each gene is itself a first-stage model estimate, the downstream confidence intervals we report treat the gene as a fixed measurement and resample only the coaches or mentors, which is the standard convention for residual- and prediction-based metrics in football analytics (Yurko et al., 2019); the reliability weighting above partly absorbs each gene's measurement noise, and a fully nested two-stage bootstrap that also resampled the gene models would be more conservative still.

3.4 Validating the Genes as Coach Traits

Before relating genes to performance, we test whether they capture stable, coach-specific tendencies rather than team or roster artifacts. We decompose the variance of each gene into the share attributable to coach identity, team identity, and season (one-way η^2). As a sharper test that exploits coaching changes, we examine the coaches who moved between teams: if a gene is a coach trait, a coach's gene at a new team should track their gene at the old team. We quantify this with the across-team correlation of a coach's old-team and new-team mean gene, and with a mover regression of the new-team gene on (a) the coach's prior-team gene and (b) the new team's pre-arrival gene, which separates the part of a coach's behavior that travels with them from the part that reflects the destination's prior identity. Finally, as a face-validity check, we rank coaches by career-average gene and compare against public reputation.

3.5 Performance Analysis

To test whether coaching genes predict performance, we correlate each composite gene with annual WAR. For the offensive aggression gene, which has the strongest theoretical prior for predicting performance, we additionally run component-level analyses and examine temporal evolution by splitting the sample into three eras (2006-2011, 2012-2017, 2018-2024), with boundaries informed by a Chow test for structural breaks.

Single-season WAR is a noisy estimate, dominated by the game-to-game variance of a 16-game season (Author, 2026). Because classical noise in the dependent variable attenuates a correlation rather than inflating it, our season-level gene-WAR correlations are conservative lower bounds, not overstatements. We address this three ways: (i) we report the relationship at the *career* level (one observation per coach, where mean WAR is far more reliable) as the primary anchor; (ii) we inverse-variance weight season-level coach-years by a games-based precision proxy; and (iii) we confirm robustness to dropping partial (sub-half) seasons, the least reliable WAR estimates. We deliberately do not report a single "disattenuated" correlation, because dividing by so low a reliability would manufacture false precision.

Because each head coach contributes many coach-years, the observations are not independent. All primary repeated-measures analyses cover the gene-WAR correlations, year-to-year persistence, mentor-protégé inheritance, and the multiple-gene regressions. For these we report *coach-clustered* inference, or *mentor-clustered* inference for mentor-protégé pairs. P-values and 95% confidence intervals are obtained from a cluster bootstrap that resamples whole coaches (or mentors) over 2,000 replicates, and the regressions additionally use cluster-robust standard errors. This is materially more conservative than treating coach-years as independent, and we treat the clustered p-value as primary throughout; conventional p-values are reported alongside where helpful. For subgroups with fewer than 40 clusters, where both the percentile cluster bootstrap and the clustered t are known to be anti-conservative, we instead report a wild cluster bootstrap (Cameron et al., 2008) (restricted, with Rademacher weights) as the more reliable inference and flag the subgroup; era-stratified and other small cross-sections are interpreted cautiously. The confidence intervals for the upstream predictive-model performance metrics (Table 3) are computed differently, by the Nadeau-Bengio (2003) corrected resampled-t: over $J = 50$ random train-test resamples that hold out whole team-years, we inflate the naive resample variance by the correction $(1/J + \rho)$ with $\rho = 0.2$ to account for the overlap among the resampled training sets, which a plain resample standard deviation badly understates (Nadeau and Bengio, 2003).

We analyze offensive and defensive coaches separately to test whether the offensive aggression-performance relationship differs by coaching background. Since all our offensive aggression measures focus on offensive decisions, we hypothesize that offensive coaches should show stronger relationships with performance.

3.6 Persistence Analysis

To measure the stability of offensive aggression over time, we construct lagged datasets matching each coach-year with their offensive aggression in subsequent years:

$$\rho_k = \mathrm{Corr}(\mathrm{Aggression}_t, \mathrm{Aggression}_{t+k}) \tag{5}$$

for lags $k \in \{1, 2, 3\}$. We estimate these correlations separately for offensive and defensive coaches to test whether coaching background affects the stability of aggressive tendencies.

3.7 Inheritance Analysis

To test whether coaching DNA transmits through coaching relationships, we match coaches with their documented mentors based on career overlap. We test inheritance for all four genes through two complementary approaches: career-average mentor-protégé correlations for offensive aggression and shotgun, where mentor-protégé pairs are available, analyzed separately by mentor background (offensive vs defensive) and coordinator type (OC vs DC); and direct coordinator-to-head-coach transitions (below), which cover all four genes, including the team-level defensive aggression gene.

For each mentor-protégé pair, we calculate average career gene scores (pooling across all seasons to reduce measurement error) and estimate:

$$\rho_{\text{inherit}} = \mathrm{Corr}(\mathrm{Gene}_{\text{protégé}}, \mathrm{Gene}_{\text{mentor}}) \tag{6}$$

Under an inheritance hypothesis, we would expect substantial positive correlations. We additionally test whether mentor WAR predicts protégé WAR, directly addressing whether working under a successful coach predicts future success.

The mentor-protégé analysis above tests inheritance through career-average offensive aggression across all coaching tree relationships. To conduct a more direct test of strategic transmission, we analyze coordinator-to-head-coach transitions specifically. Using complete career histories from Pro Football Reference, we identify every coach who served as a defensive coordinator (DC) before becoming a head coach, and every coach who served as an offensive coordinator (OC) before becoming a head coach.

For DC→HC transitions, we compare the team's defensive aggression gene during the coordinator's DC tenure with the team's defensive aggression gene during their HC tenure. For OC→HC transitions, we compare the team's offensive genes (offensive aggression, shotgun, tempo) during the OC's coordinator tenure with their HC-era genes. When a coach had multiple coordinator stints, we keep a single transition per coach and role: their most recent coordinator stint that precedes their head-coaching tenure, with the gene averaged over that stint's seasons. This uses all of the stint's data while avoiding pseudo-replication of the HC-era outcome. Coordinator years must precede HC years to ensure proper temporal ordering.

This approach tests a more specific hypothesis than the career-average analysis: does the *team-level* strategic identity that a coordinator helped create persist when they take the helm of their own team?

3.8 Multiple Comparison Correction

Our analysis involves numerous hypothesis tests across gene-WAR performance relationships, offensive aggression component analyses, persistence, inheritance patterns, within-coach causal effects, and temporal robustness. To address concerns about false discoveries from multiple testing, we apply the Benjamini-Hochberg procedure to control the False Discovery Rate (FDR) at $\alpha = 0.05$ (Benjamini and Hochberg, 1995).

We apply a single global Benjamini-Hochberg family spanning all 120 hypothesis tests in the study, preferring the wild cluster bootstrap p-value for small-cluster subgroups, the percentile clustered p-value for other repeated-measures tests, and the conventional p-value otherwise. This is the most conservative, least forking-path-prone choice. Under this family the Benjamini-Hochberg threshold falls at $p \leq 0.024$, below which 69 of the 120 tests lie. We treat this threshold as one decision rule rather than a sharp boundary and report both clustered and conventional p-values throughout.

3.9 Within-Coach Fixed Effects

To strengthen causal inference, we employ two-way fixed effects regression that controls for both coach-specific quality and temporal changes in offensive aggression's effectiveness. This approach isolates whether coaches perform better when they are more aggressive than usual, removing confounds from stable coach characteristics and league-wide temporal trends.

The model specification is:

$$\text{WAR}_{it} = \beta \cdot \text{Aggression}_{it} + \alpha_i + \gamma_t + \epsilon_{it} \tag{7}$$

where α_i represents coach fixed effects (controlling for time-invariant coach quality) and γ_t represents year fixed effects (controlling for temporal changes in the value of offensive aggression). We demean variables within both coaches and years, then estimate β using OLS with standard errors clustered by coach.

This two-way fixed effects approach is critical because coaches' offensive aggression may increase over time as league-wide norms shift. Without year fixed effects, within-coach increases in offensive aggression would be confounded with the declining competitive advantage of offensive aggression, potentially obscuring a genuine causal relationship. The two-way approach asks: when a coach is more aggressive than the league average in year t (relative to their typical deviation from league norms), do they perform better?

4 Results

4.1 Coaching Genes Are Coach Traits

Before asking whether genes predict performance, we establish that they measure something real and coach-specific, a stable strategic identity that belongs to the coach rather than to the team or its roster. Three lines of evidence converge.

A variance decomposition shows that, across every gene, coach identity explains far more variance than team identity. For composite offensive aggression, coach identity accounts for 51% of the variance against 13% for team; the same ordering holds for every gene and component (Table 4). A coach's genes are roughly four to five times more a function of who they are than of where they coach.

The gene also travels with the coach, a sharper test that exploits coaching changes. Among the 32 coaches who changed teams during our window, the gene a coach exhibits at a new team tracks the gene they exhibited at their previous team, with cross-team correlations comparable to or stronger than the year-to-year persistence *within* a single team (Table 4; Section 4.6). A mover regression refines the attribution. Regressing a coach's new-team gene on both their prior-team gene and the new team's pre-arrival gene, the coach term is large and well-determined (composite offensive aggression $\beta = 0.53$, 95% CI $[0.26, 0.83]$, $p < 0.001$), while the new team's pre-arrival tendency also carries weight ($\beta = 0.72$, 95% CI $[0.21, 1.14]$, $p = 0.004$), consistent with teams hiring coaches whose style fits the building. The strong cross-team travel of the gene nonetheless shows that strategic identity is substantially carried by the coach.

Tab. 4: Genes Are Coach Traits: Variance Decomposition and Cross-Team Travel

Gene	Var. Coach	Var. Team	Var. Season	Travel r [95% CI]
Composite Offensive Aggression	0.51	0.13	0.19	0.61 [0.33, 0.78]
Shotgun Formation	0.58	0.12	0.49	0.55 [0.29, 0.75]
Tempo	0.53	0.10	0.08	0.54 [−0.11, 0.86]
4th Down	0.53	0.06	0.28	0.18 [−0.25, 0.51]
Pass-Heavy	0.49	0.19	0.08	0.42 [0.16, 0.65]
Deep Pass	0.44	0.12	0.06	0.16 [−0.20, 0.50]
Two-Point	0.44	0.10	0.17	0.36 [−0.02, 0.69]

Variance shares are one-way η^2 for coach, team, and season identity, each computed marginally, so they need not sum to one and carry no CI (coach and team are confounded). Travel r is the correlation of a coach's old-team vs. new-team mean gene across the 32 movers, with a coach bootstrap 95% CI.

The composite travels better than any single component, because aggregating the four facets averages out their component-level sampling noise. Among components, pass-heavy and two-point tendencies travel moderately while fourth-down and deep-pass travel weakly, indicating that the broad disposition is more portable than any one decision rule.

The genes also have face validity, recovering coaches' public reputations. The most aggressive coaches by career-average composite are Doug Pederson, Brandon Staley, and Dirk Koetter, with Bruce Arians just behind; the least are Dick Jauron, Brad Childress, and John Fox. Tempo is led by Chip Kelly and Kliff Kingsbury, shotgun by Nick Sirianni and Chip Kelly, and fourth-down aggression by Dan Campbell. Standardizing each career-average gene across coaches (Table 5) shows these leaders sit two to three standard deviations from the average coach (Chip Kelly's tempo, an extreme outlier, sits near +7), and in each case the coaches the measure flags are those a knowledgeable observer would name.

Tab. 5: Face Validity: Career-Average Gene Leaders and Laggards (z-scored)

Gene	Most (z)	Least (z)
Comp. Off. Aggression	Pederson (+2.7), Staley (+2.6), Koetter (+2.1)	Jauron (−2.3), Childress (−2.0), Fox (−1.6)
4th Down	Campbell (+3.7), LaFleur (+2.8), Staley (+2.7)	Schwartz (−1.4), McCoy (−1.3), Dungy (−1.3)
Shotgun	Sirianni (+2.5), Kingsbury (+2.3), Kelly (+2.2)	Holmgren (−3.1), Nolan (−2.3), Marinelli (−1.9)
Tempo	Kelly (+6.8), Kingsbury (+3.0), Spagnuolo (+1.6)	Nolan (−1.4), LaFleur (−1.3), Gailey (−1.2)

Career-average gene, coaches with ≥ 3 seasons ($n = 89$), standardized across coaches (z = standard deviations from the average qualifying coach).

Having established that the genes are stable, coach-specific traits, we now ask what they predict.

4.2 Coaching Genes and Performance

We first test whether our four gene dimensions predict coaching performance. Because single-season WAR is dominated by the game-to-game luck of a 16-game season, we anchor on the *career* level, correlating each coach's career-average gene with their career-average WAR, where the outcome is far more reliable. Table 6 reports these correlations.

Tab. 6: Coaching Genes vs. Career-Average WAR

Gene	r	95% CI	p	Coaches
Composite Offensive Aggression	0.303	[0.16, 0.45]	<0.001	124
Shotgun Formation	0.230	[0.04, 0.41]	0.010	124
Defensive Aggression	0.188	[0.02, 0.35]	0.10	79
Composite Tempo	0.058	[−0.06, 0.22]	0.52	124

Correlation of each coach's career-average gene with career-average WAR (one observation per coach), with a coach bootstrap p and 95% CI. Defensive aggression covers 2016–2024; all others 2006–2024. Season-level coach-year correlations agree in sign (Table 13).

Three of the four genes are positively associated with career performance. Composite offensive aggression shows the strongest relationship ($r = 0.303$, 95% CI $[0.16, 0.45]$, $p < 0.001$, $n = 124$ coaches), followed by shotgun formation preference ($r = 0.230$, $[0.04, 0.41]$, $p = 0.010$) and defensive aggression ($r = 0.188$, $[0.02, 0.35]$, $p = 0.10$, $n = 79$), the last positive but imprecise given its shorter 2016–2024 window. Tempo shows essentially no relationship ($r = 0.058$, $[-0.06, 0.22]$, $p = 0.52$).

Our temporal analyses below necessarily use season-level coach-years, where the corresponding correlations agree in sign with the career anchor and are robust to inverse-variance weighting by WAR precision and to dropping partial seasons (composite offensive aggression $r = 0.21$ in both checks). Given that most coaching genes carry real performance signal, we turn to whether they transmit through coaching relationships, and whether the genes that transfer most strongly are the same ones that predict winning.

4.3 Testing the Theory of Coaching Trees

The coaching tree hypothesis makes three testable claims: that successful mentors produce successful protégés, that strategic tendencies transmit through apprenticeship, and that coaching DNA persists through promotion. We test each in turn.

4.3.1 Performance Inheritance: Does Mentor Success Predict Protégé Success?

The most fundamental claim of the coaching tree narrative is that working under a great coach produces great coaches. We test this by correlating average career WAR for mentors with average career WAR for their protégés across 207 documented modern-era mentor-protégé pairs where both served as head coaches.

Tab. 7: Mentor WAR vs. Protégé WAR

Relationship	r	95% CI	p	n
Overall	0.124	[−0.01, 0.26]	0.074	192
OC → HC	0.087	[−0.13, 0.30]	0.45	101
DC → HC	0.190	[0.01, 0.37]	0.045	89

Mentor-clustered bootstrap p and 95% CI

Mentor performance shows at most a weak association with protégé performance ($r = 0.124$, 95% CI $[-0.01, 0.26]$, $p = 0.074$), explaining under 2% of the variance. The association is negligible for offensive coordinator protégés ($r = 0.087$, $[-0.13, 0.30]$, $p = 0.45$) and only modest for defensive coordinator protégés

($r = 0.190$, $[0.01, 0.37]$, $p = 0.045$), an estimate whose interval still reaches near zero. On average, the data give little indication that working under a highly successful coach, rather than an average one, predicts greater success as a head coach.

4.3.2 Offensive Aggression Inheritance

If coaching trees don't transmit success, do they transmit strategy? We analyze 513 documented mentor-protégé pairs where we have offensive aggression data for both, using career average offensive aggression scores to maximize reliability. Table 8 presents results.

Tab. 8: Offensive Aggression Inheritance: Overall Mentor-Protégé Pairs

Component	r	95% CI	P-value
Composite	0.080	[−0.05, 0.22]	0.24
4th Down	0.172	[−0.01, 0.34]	0.07
Pass-Heavy	0.025	[−0.13, 0.20]	0.69
Deep Pass	0.085	[−0.07, 0.23]	0.27
Two-Point	0.050	[−0.07, 0.18]	0.36

N=513 mentor-protégé pairs; mentor-clustered p and 95% CI
Every CI includes zero after clustering

Composite offensive aggression shows little inheritance ($r = 0.080$, 95% CI $[-0.05, 0.22]$, $p = 0.24$). A mentor's offensive aggression carries little information about the protégé's. Fourth-down decision-making is the strongest component ($r = 0.172$, $[-0.01, 0.34]$, $p = 0.07$), and is more pronounced under conventional unclustered inference, consistent with our persistence findings that 4th-down philosophy is the most stable, trait-like component; once the repeated appearance of the same mentors is taken into account, however, the signal is weak.

Separating by coordinator type is informative. Table 9 shows that among OC-mentored protégés, pass-heavy tendency shows the clearest inheritance ($r = 0.333$, 95% CI $[0.11, 0.51]$, clustered $p = 0.004$), while the 4th-down association ($r = 0.232$, $[-0.05, 0.49]$) is weaker once mentor clustering is applied ($p = 0.12$). DC-mentored protégés show little inheritance for any offensive aggression component.

Tab. 9: Offensive Aggression Inheritance by Coordinator Type

	OC → HC		DC → HC	
Component	r [95% CI]	p	r [95% CI]	p
Composite	0.218 [−0.01, 0.44]	0.07	0.019 [−0.21, 0.27]	0.89
4th Down	0.232 [−0.05, 0.49]	0.12	0.016 [−0.16, 0.18]	0.86
Pass-Heavy	0.333 [0.11, 0.51]	0.004	−0.040 [−0.28, 0.22]	0.75
Deep Pass	0.101 [−0.12, 0.28]	0.35	0.094 [−0.12, 0.30]	0.38
Two-Point	0.060 [−0.14, 0.29]	0.52	0.048 [−0.09, 0.19]	0.48

OC: $n = 200$ (40 mentors), DC: $n = 232$ (33 mentors); mentor-clustered p and 95% CI
DC p-values are wild cluster bootstrap (fewer than 40 mentors)

4.3.3 Shotgun Formation Inheritance

The offensive aggression results suggest weak-to-absent strategic transmission. But offensive aggression captures situational decision-making, the in-the-moment choices about risk. What about schematic identity, the structural elements of how a team looks pre-snap? The shotgun formation gene provides a direct test.

Tab. 10: Shotgun Gene Inheritance: Mentor-Protégé Pairs

Relationship	r	95% CI	p	n
Overall	0.289	[0.16, 0.41]	<0.001	608
OC → HC	0.399	[0.21, 0.54]	<0.001	249
DC → HC	0.096	[−0.12, 0.30]	0.43	269

Mentor-clustered bootstrap p and 95% CI

Shotgun formation preference shows clearer overall inheritance ($r = 0.289$, 95% CI $[0.16, 0.41]$, $p < 0.001$, $n = 608$), several times stronger than composite offensive aggression ($r = 0.080$, $[-0.05, 0.22]$), and the association holds under mentor clustering.

Shotgun inheritance is specific to offensive backgrounds. Offensive coordinator protégés show clear inheritance ($r = 0.399$, 95% CI $[0.21, 0.54]$, $p < 0.001$), while defensive coordinator protégés show little ($r = 0.096$, $[-0.12, 0.30]$, $p = 0.43$). This domain-specificity is consistent with the mechanism: offensive coordinators design and install formation packages, while defensive coordinators have no direct role in shotgun usage decisions.

4.3.4 Synthesis: What Coaching Trees Actually Transmit

Table 11 synthesizes the coaching tree findings across all dimensions tested.

Tab. 11: Summary: What Coaching Trees Transmit

Dimension	Type	r	95% CI	Transmitted?
Performance (WAR)	Outcome	0.124	[−0.01, 0.26]	No
Composite Offensive Aggression	Situational	0.080	[−0.05, 0.22]	No
4th Down Aggression	Situational	0.172	[−0.01, 0.34]	No†
Shotgun Formation (OC)	Schematic	0.399	[0.21, 0.54]	Strongly

Mentor-clustered; Shotgun DC → HC: $r = 0.096$, near zero
† Stronger under unclustered p; weak after clustering ($p = 0.08$)

The pattern across dimensions is consistent. Coaching trees show little transmission of performance or general risk tolerance, but offensive coaching trees *do* transmit schematic identity, the structural elements of how a team looks on the field. The distinction maps cleanly onto the difference between system design (transmissible) and situational judgment (personal), with the further finding that transmission is domain-specific. Only coaches who designed the system transmit it.

Figure 1 visualizes the offensive aggression inheritance correlations by coordinator type.

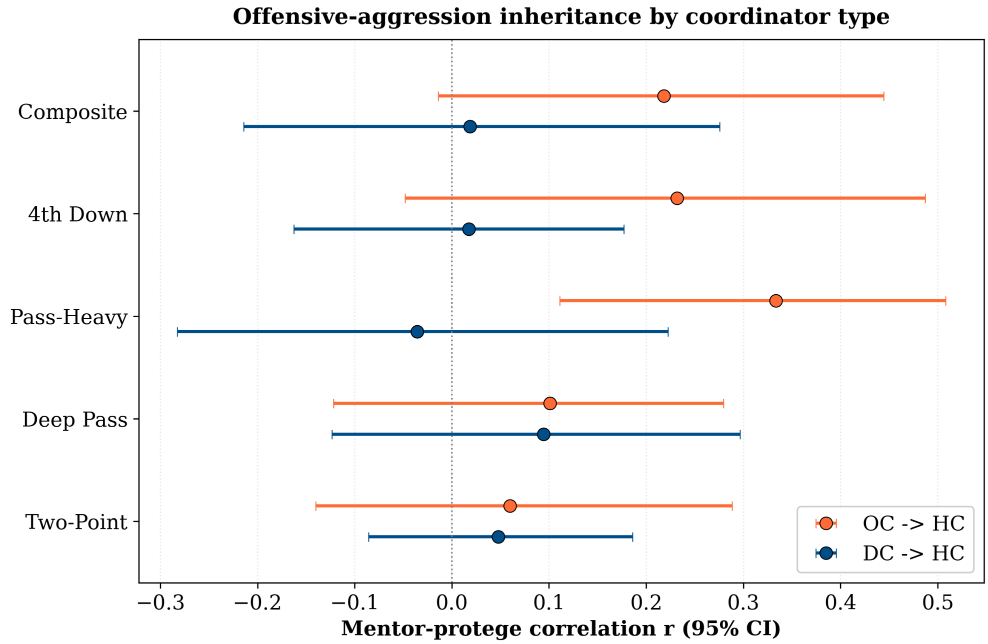

Fig. 1: Offensive-aggression inheritance by coordinator type: the mentor-protégé correlation r (with 95% CI) for each component, comparing OC→HC (orange) and DC→HC (blue) transitions. OC inheritance is positive across components, clearest for the pass-heavy tendency, while DC inheritance is near zero throughout.

4.4 Direct Coordinator-to-HC Gene Transmission

The mentor-protégé analysis above tests inheritance through career averages across all coaching tree relationships. We now conduct a more direct test: when a coordinator becomes a head coach, does the team they lead resemble the team they previously coordinated? Using complete career histories, we identify every DC→HC and OC→HC transition and compare team-level gene values during the coordinator era directly to the HC era. When a coach had multiple coordinator stints, we keep one transition per coach: their most recent coordinator stint preceding the head-coaching role, with the gene averaged over that stint's seasons, which avoids pseudo-replication of the HC-era outcome while using all of the stint's data.

Table 12 presents the results across all four gene dimensions.

Tab. 12: Coordinator-to-HC Gene Inheritance

Gene	Transition	n	r	95% CI	p	Dir. Ret.
Defensive Aggression	DC→HC	17	0.570	[0.22, 0.82]	0.020	65%
Shotgun	OC→HC	40	0.674	[0.48, 0.80]	<0.001	82%
Tempo	OC→HC	40	0.318	[−0.15, 0.64]	0.23	58%
Offensive Aggression	OC→HC	40	−0.008	[−0.34, 0.29]	0.93	52%

Dir. Ret. = percentage of coaches whose gene retains the same sign. One transition per coach (the most recent coordinator stint preceding the HC role, averaged over its seasons).
95% CIs are the percentile coach cluster bootstrap. Defensive aggression ($n = 17$) uses a wild cluster bootstrap p; the offensive transitions ($n = 40$) use the coach-clustered bootstrap p.

The results reinforce the mentor-protégé findings. Shotgun formation preference, the most visible schematic element of offensive identity, shows strong direct transmission ($r = 0.674$, 95% CI $[0.48, 0.80]$, $p < 0.001$), consistent with the OC mentor-protégé correlation ($r = 0.399$, Table 10). When an offensive coordinator runs shotgun-heavy formations, about 82% of the time their subsequent head-coaching tenure shows the same tendency. Matt Nagy, for instance, carried a shotgun-heavy identity from his Kansas City coordinator years to his Chicago head-coaching tenure, and Jason Garrett a below-average one from his Dallas coordinator years through his decade as the team's head coach.

Defensive aggression also shows a positive association ($r = 0.570$, 95% CI $[0.22, 0.82]$; $p = 0.020$ from a wild cluster bootstrap appropriate to the small sample), though with only $n = 17$ transitions, limited by defensive tracking data from 2016 onward, this is suggestive rather than definitive. Dan Quinn, for example, carried an aggressive defense from his Dallas coordinator tenure to Washington, and Matt Patricia a conservative one from New England to Detroit.

Tempo shows a moderate point estimate that is not statistically reliable at this sample size ($r = 0.318$, 95% CI $[-0.15, 0.64]$; coach-clustered bootstrap $p = 0.23$, naive $p = 0.046$), so we do not read it as dependable coordinator-to-HC inheritance. Offensive Aggression shows essentially none ($r = -0.008$, $[-0.34, 0.29]$, $p = 0.93$), consistent with the career-average analysis. Among the four, shotgun and defensive aggression are the strongest and most precisely estimated transitions, while tempo and offensive aggression are weak.

Figure 2 visualizes these patterns, contrasting shotgun (positive slope, tight clustering) with offensive aggression (flat, dispersed).

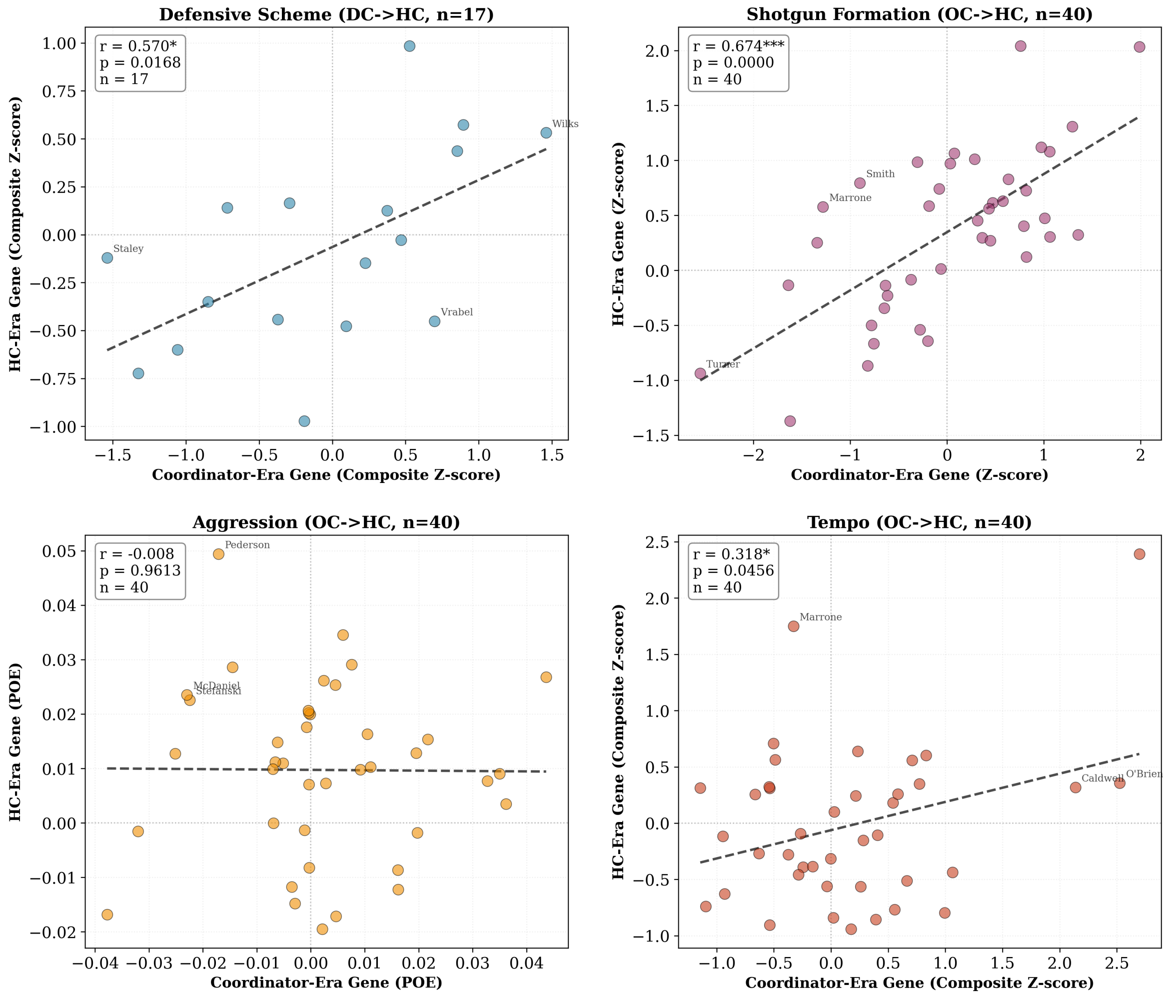

Fig. 2: Coordinator-era gene versus HC-era gene for all four dimensions. Each point represents one coordinator-to-HC transition. Shotgun formation and defensive aggression show strong positive correlations, indicating that schematic identity persists. Offensive Aggression shows no relationship and tempo only a weak, statistically unreliable one, indicating that situational decision-making and pace do not reliably transfer.

Figure 3 summarizes the inheritance strength and direction retention across all four genes.

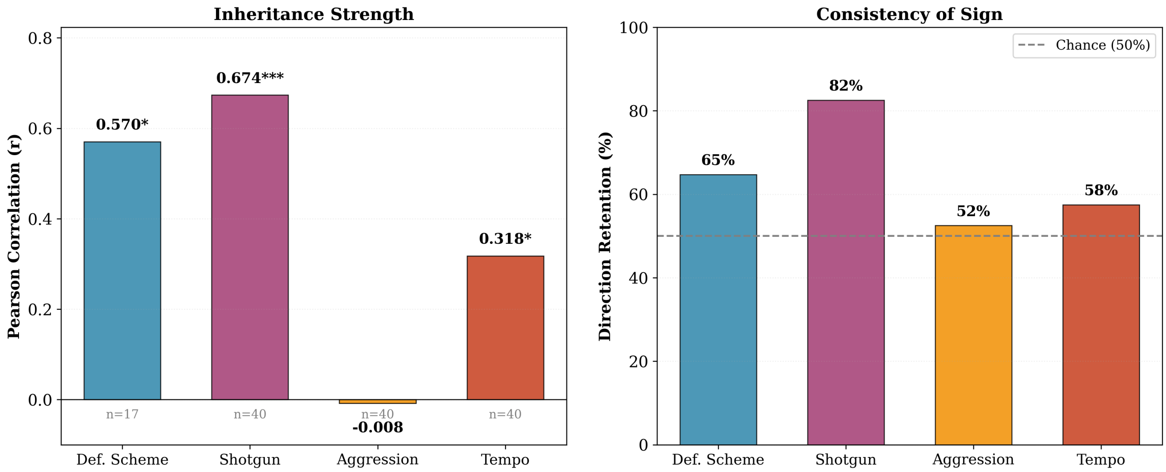

Fig. 3: Left: Pearson correlation between coordinator-era and HC-era genes. Right: Percentage of coaches whose gene retains the same sign (positive/negative) across transitions. The dashed line indicates chance level (50%). Schematic genes (defensive aggression, shotgun) show strong inheritance while situational decision-making (offensive aggression) and tempo do not.

4.5 Offensive Aggression and Performance: A Disappearing Advantage

At the season level, where our temporal tests operate, the composite offensive aggression-WAR relationship masks a dynamic story. Table 13 presents the component relationships pooled across 2006-2024.

Tab. 13: Offensive Aggression vs. WAR: Overall (2006-2024)

Measure	r	95% CI	P-value
Composite	0.189	[0.10, 0.28]	<0.001
Pass-Heavy	0.183	[0.07, 0.28]	0.002
4th Down	0.095	[0.01, 0.18]	0.033
Deep Pass	0.020	[−0.07, 0.11]	0.65
2-Point	0.037	[−0.06, 0.13]	0.45
N=608 coach-years; coach-clustered bootstrap p and 95% CI			

Composite offensive aggression is positively related to performance ($r = 0.189$, 95% CI $[0.10, 0.28]$, $p < 0.001$), driven primarily by the pass-heavy component ($r = 0.183$, $[0.07, 0.28]$, $p = 0.002$); the fourth-down component is positive but weaker ($r = 0.095$, $[0.01, 0.18]$, $p = 0.033$), and stronger at the career level ($r = 0.187$). Splitting the sample into three eras (2006-2011, $n = 192$; 2012-2017, $n = 192$; 2018-2024, $n = 224$) reveals a decline (Table 14, Figure 4).

Tab. 14: Offensive Aggression vs. WAR by Era

Era	Composite (r [95% CI], p)	Pass-Heavy (r [95% CI], p)	N
2006-2011	0.239 [0.07, 0.38], 0.001	0.173 [0.00, 0.32], 0.016	192
2012-2017	0.137 [−0.07, 0.29], 0.059	0.175 [0.01, 0.32], 0.015	192
2018-2024	0.101 [−0.03, 0.25], 0.132	0.131 [−0.04, 0.31], 0.049	224
Unclustered subgroup p; 95% CIs are coach-clustered bootstrap			

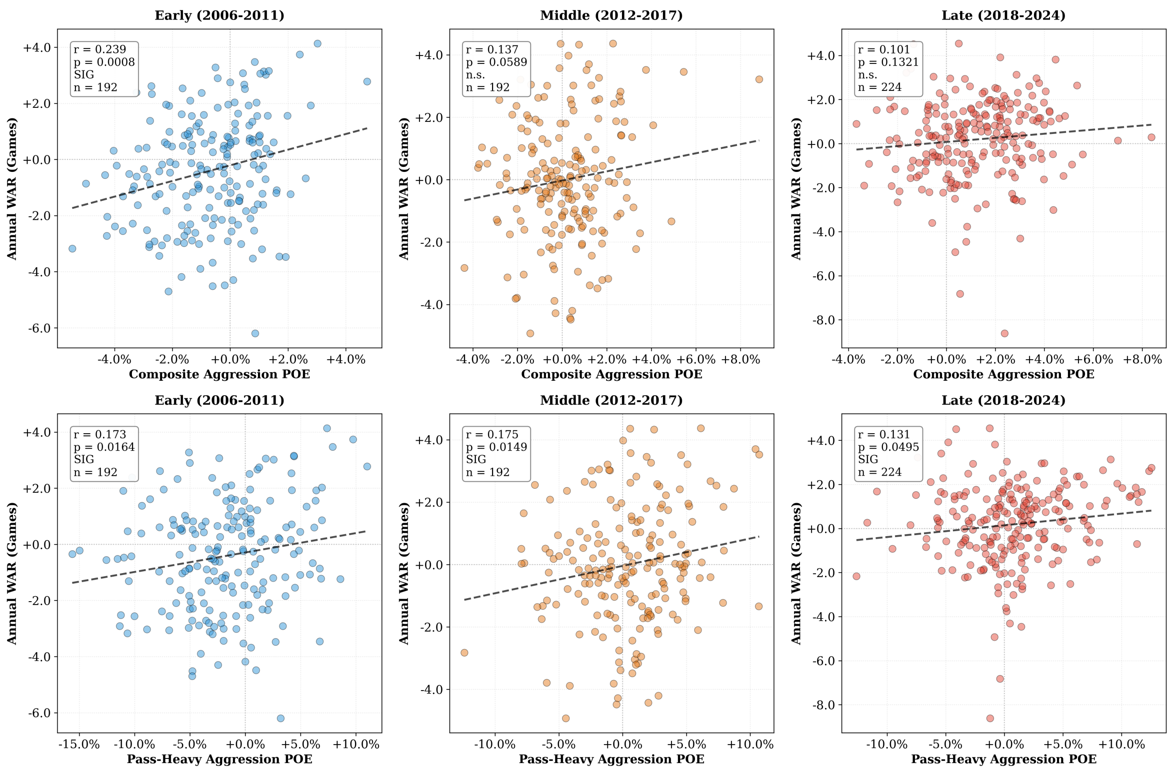

Fig. 4: Scatterplots of composite offensive aggression (top row) and pass-heavy aggression (bottom row) versus annual WAR, split by era. The strength of the composite offensive aggression relationship has declined from $r = 0.239$ ($p = 0.001$) in the early era to $r = 0.101$ ($p = 0.13$) in the late era, while the pass-heavy component remains positive throughout.

Composite offensive aggression's edge roughly halved across the window, from $r = 0.239$ (95% CI $[0.07, 0.38]$, $p = 0.001$) in the early era to $r = 0.137$ ($[-0.07, 0.29]$, $p = 0.059$) in the middle and $r = 0.101$ ($[-0.03, 0.25]$, $p = 0.13$) in the modern era. The advantage diminished rather than vanished: the pass-heavy component in particular remains positive even in the modern era ($r = 0.131$, $[-0.04, 0.31]$, $p = 0.049$). This erosion coincides with a leaguewide shift in the tactic itself (Figure 5): mean composite offensive aggression climbed from below the model's season-agnostic baseline early, crossing it around 2013, to clearly above it by the modern era, and its spread across coaches widened by about a fifth, the diffusion pattern one expects as an edge becomes common knowledge.

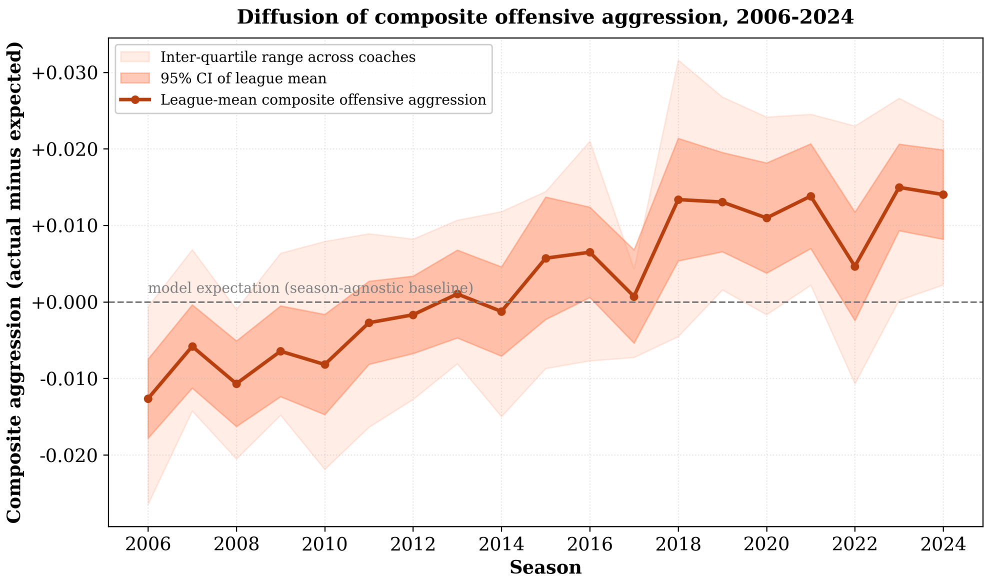

Fig. 5: Diffusion of composite offensive aggression, 2006–2024. The league-mean composite offensive aggression gene (actual minus the season-agnostic model expectation, reliability-weighted across the four sub-components; dark line, 95% CI shaded) rises from below the time-invariant baseline in the early era, crossing it around 2013, to clearly above it by the modern era, while the inter-quartile range across coaches (light band) widens. Because the baseline model excludes season, the upward drift is the adoption signal: average aggression rose and coaches spread out in how far they pushed it.

4.5.1 Temporal Robustness

To probe whether the erosion is genuine, we conduct two complementary tests. A continuous-year interaction model (WAR $= \beta_0 + \beta_1 \cdot$ Aggression $+ \beta_2 \cdot$ Year $+ \beta_3 \cdot$ Aggression $\times$ Year, with Year centered at 2015 and coach-clustered standard errors) yields $\beta_3 = -0.88$ ($SE = 0.74$, 95% CI $[-2.36, 0.59]$, $p = 0.24$). The point estimate is negative, consistent with decline, but its confidence interval includes zero, so the continuous test alone does not establish erosion. The implied effect nonetheless falls from $\beta = 25.1$ (95% CI $[5.7, 44.6]$, $p = 0.012$) in 2006 to $\beta = 9.2$ ($[-4.9, 23.3]$, $p = 0.20$) in 2024, fading toward zero rather than reversing sign. The firmer evidence comes from structural-break testing. A Chow test points to a break at 2012 ($F = 3.41$, $p = 0.034$), with the offensive aggression coefficient falling from $\beta = 27.2$ (95% CI $[9.6, 43.8]$) before the break to $\beta = 10.7$ ($[0.2, 20.4]$) after, a 61% reduction; a break at 2013 is comparable ($p = 0.038$). Both lie just above the strict global false-discovery threshold, so we read them as suggestive. Taken with the era pattern, this points to a real but partial erosion rather than a clean disappearance. An out-of-sample check is consistent with this reading. Fitting the season-level relationship on 2006–2018 and applying it to 2019–2024, the offensive aggression-WAR relationship generalizes only weakly (test $r = 0.12$), as expected when the outcome is dominated by single-season luck, while the defensive-aggression-WAR relationship is concentrated in the most recent seasons (near zero before 2019, $r = 0.25$ after).

4.5.2 Offensive vs. Defensive Coaches

We next ask whether the gene-WAR relationship depends on a coach's background, separating offensive-minded from defensive-minded head coaches for both the offensive and defensive aggression genes (Table 15).

Tab. 15: Offensive and Defensive Aggression vs. WAR by Coach Background

	Offensive coaches		Defensive coaches	
Gene	**r [95% CI]**	**p**	**r [95% CI]**	**p**
Offensive Aggression	0.211 [0.09, 0.33]	0.002	0.184 [0.03, 0.33]	0.016
Defensive Aggression	0.154 [0.01, 0.30]	0.048	0.027 [−0.14, 0.21]	0.78

Coach-clustered bootstrap p and 95% CI. Offensive aggression: $n = 310$ offensive / 251 defensive coach-years (2006–2024); defensive aggression: $n = 160$ / 100 (2016–2024). The defensive-coach defensive-aggression cell (< 40 coaches) uses a wild cluster bootstrap p.

For offensive aggression, both backgrounds show positive relationships of similar magnitude with WAR (offensive coaches $r = 0.211$, 95% CI $[0.09, 0.33]$, $p = 0.002$; defensive coaches $r = 0.184$, $[0.03, 0.33]$, $p = 0.016$), so situational offensive aggression relates to performance regardless of which side of the ball a coach comes from. Defensive aggression behaves differently: it is positively associated with WAR for offensive-minded head coaches ($r = 0.154$, $[0.01, 0.30]$, $p = 0.048$, above the global false-discovery threshold) but shows essentially no relationship for defensive-minded ones ($r = 0.027$, $[-0.14, 0.21]$, wild cluster $p = 0.78$). The defensive-coach estimate rests on only 29 coaches and is imprecise, so we read this contrast cautiously; it nonetheless runs counter to the intuition that defensive specialists would gain the most from an aggressive scheme.

4.5.3 Within-Coach Fixed Effects: Causal Evidence

Does offensive aggression cause better performance, or do better coaches simply become more aggressive? To address this, we employ two-way fixed effects regression controlling for both coach quality and temporal trends. The within-coach approach compares each coach to themselves across different years, isolating behavioral effects from selection confounds.

Table 16 presents the results. A two-way fixed-effects specification that demeans offensive aggression within both coach and year indicates a positive within-coach effect ($\beta = 1.40$, 95% CI $[0.76, 1.96]$, $p < 0.001$). Being more aggressive than one's own norm is associated with better performance, which is not simply a matter of better coaches happening to be more aggressive. A cluster-robust effect-size analysis quantifies the magnitude. A one standard deviation increase in offensive aggression corresponds to 0.44 additional WAR games (Cohen's $d = 0.23$), about 17% of the IQR in the WAR distribution (2.512 games), with 99% power in the pooled sample.

Tab. 16: Within-Coach Fixed Effects: Offensive Aggression → WAR

Sample	**Within-coach β**	**95% CI**	**p**	**n**
Pooled (2006-2024)	1.40	[0.76, 1.96]	<0.001	587
Early (2006-2011)	1.72	[0.30, 3.05]	0.019	177
Middle (2012-2017)	0.78	[−0.41, 1.78]	0.160	180
Late (2018-2024)	1.18	[0.06, 2.09]	0.024	207

Two-way fixed effects (offensive aggression demeaned within coach and year), coach-clustered SEs with a coach-block bootstrap 95% CI. Pooled effect-size translation: 0.44 WAR-games per SD (Cohen's $d = 0.23$, 17% of the 2.512-game IQR, 99% power).

Era-stratified within-coach estimates do not follow a monotone trend ($\beta = 1.72$, 0.78, 1.18; the early and late eras are individually significant while the middle era is not; 95% CIs in Table 16), so they neither confirm nor rule out erosion of the within-coach effect. The pooled analysis provides the most reliable estimate (99% power).

4.6 Offensive Aggression as Trait and State

Understanding whether offensive aggression represents stable individual traits or adaptive responses to circumstances has important implications for talent evaluation. Our persistence analysis examines year-to-year correlations in coaching behavior.

Table 17 presents the year-over-year (lag 1) correlations for each offensive aggression component, based on 467 consecutive coach-year pairs where the same coach was in place for two consecutive seasons. Note that these estimates may be upward-biased due to survivorship: coaches who retain their positions across multiple seasons are disproportionately successful, and successful coaches may have more stable situations that enable consistent strategic approaches.

Tab. 17: Year-to-Year Offensive Aggression Persistence (Lag 1)

Component	r	95% CI	R^2
Composite	0.516	[0.42, 0.60]	0.266
4th Down	0.524	[0.42, 0.61]	0.274
Pass-Heavy	0.552	[0.45, 0.64]	0.304
Deep Pass	0.472	[0.35, 0.57]	0.223
2-Point	0.400	[0.26, 0.53]	0.160

Coach-clustered p and 95% CI; all clustered $p < 0.001$

All five measures show substantial persistence (clustered $p < 0.001$ throughout). The correlations range from 0.40 to 0.55, indicating that offensive aggression in year N predicts 16-30% of the variance in year N+1 offensive aggression. This is substantial enough to constitute a real trait, yet modest enough to allow for year-to-year variation from roster changes, staff turnover, and strategic adaptation.

Pass-heavy and fourth-down tendencies are the most persistent components ($r = 0.55$, 95% CI $[0.45, 0.64]$, and $r = 0.52$, $[0.42, 0.61]$). A coach who goes for it frequently on fourth down one year is quite likely to do so again the next, even as rosters turn over and circumstances change, suggesting that fourth-down philosophy in particular reflects genuine strategic preferences that persist across contexts.

Figure 6 shows how these correlations decay as we extend the lag. At lag 2 (two years apart), correlations drop to the 0.23–0.44 range. At lag 3, they settle in the 0.31–0.40 range. Even at three years apart, all correlations remain clearly positive.

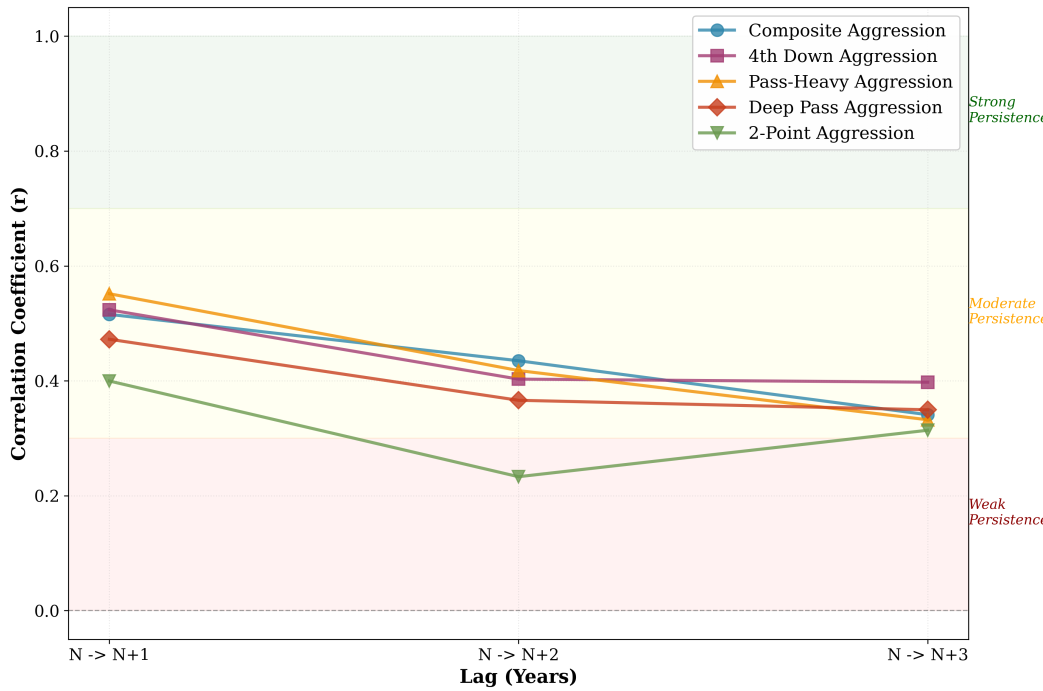

Fig. 6: Persistence of offensive aggression over 1, 2, and 3 year lags. Fourth down aggression (purple squares) shows the strongest and most stable persistence, while two-point aggression (green triangles) decays most rapidly. All measures remain clearly positive even at 3-year lags.

4.6.1 Offensive vs. Defensive Coaches

Persistence patterns differ by coaching background. Table 18 shows correlations at 1-, 2-, and 3-year lags separated by coach type.

Tab. 18: Persistence by Coach Background (Lags 1-3)

	Offensive			Defensive		
Component	**Lag 1**	**Lag 2**	**Lag 3**	**Lag 1**	**Lag 2**	**Lag 3**
Composite	0.550	0.473	0.319	0.518	0.422	0.395
	[0.41, 0.66]	[0.31, 0.61]	[0.10, 0.55]	[0.39, 0.62]	[0.27, 0.52]	[0.23, 0.52]
4th Down	0.639	0.586	0.483	0.352	0.212	0.356
	[0.52, 0.72]	[0.41, 0.71]	[0.21, 0.69]	[0.22, 0.47]	[0.00, 0.39]	[0.14, 0.54]
Pass-Heavy	0.622	0.494	0.449	0.522	0.350	0.273
	[0.47, 0.73]	[0.21, 0.64]	[0.13, 0.61]	[0.39, 0.62]	[0.16, 0.48]	[0.08, 0.48]
Two-Point	0.468	0.402	0.456	0.438	0.259	0.283
	[0.19, 0.65]	[0.16, 0.58]	[0.18, 0.67]	[0.35, 0.54]	[0.10, 0.47]	[0.11, 0.58]

Offensive: $n_{lag1} = 228$, $n_{lag2} = 160$, $n_{lag3} = 110$; Defensive: $n_{lag1} = 194$, $n_{lag2} = 150$, $n_{lag3} = 113$
Coach-clustered bootstrap p and 95% CI (bracketed); significance is read from whether a CI excludes zero

Offensive coaches show stronger year-to-year persistence than defensive coaches for the specific components, most clearly 4th down decisions ($r = 0.639$, 95% CI $[0.52, 0.72]$, vs $r = 0.352$, $[0.22, 0.47]$). The gap is substantial: offensive coaches explain 41% of next year's 4th down aggression from this year's value, while defensive coaches explain only 12%. For composite offensive aggression, however, the gap is small ($r = 0.550$,

$[0.41, 0.66]$, vs $r = 0.518$, $[0.39, 0.62]$; difference 0.032): the offensive-defensive divergence is concentrated in the 4th-down and pass-heavy components rather than in overall offensive aggression.

This pattern extends across longer lags. At 2-year lags, the 4th down persistence gap is largest ($r = 0.586$, 95% CI $[0.41, 0.71]$, vs $r = 0.212$, $[0.00, 0.39]$; difference $+0.374$). At 3-year lags, sizable gaps persist for 4th down and pass-heavy aggression, while for defensive coaches fourth-down aggression at the 2-year lag is no longer distinguishable from zero.

The interpretation is nuanced. For offensive coaches, the component tendencies, especially 4th down philosophy, behave as ingrained traits reflecting core beliefs about optimal strategy, persisting strongly across contexts. For defensive coaches making offensive decisions, those same component tendencies are more contextual and reactive, adapting to personnel, opponents, and situations. Overall composite persistence is nonetheless comparable across backgrounds, indicating both groups maintain a stable aggregate identity even as the underlying components differ in stickiness.

5 Discussion

5.1 The Erosion of a Competitive Advantage

Aggressive play-calling provided a measurable edge in 2006-2011 ($r = 0.239$, 95% CI $[0.07, 0.38]$, $p = 0.001$) that diminished to $r = 0.101$ ($[-0.03, 0.25]$, $p = 0.13$) by 2018-2024. The continuous year-by-aggression interaction is negative but imprecise ($p = 0.24$), and the 2012 structural break ($p = 0.03$) is modest on its own; together with the era pattern, this points to a partial erosion that we treat as suggestive rather than firmly established. As analytics-driven decision-making spread, aggressive tactics became more common: fourth-down aggression in particular moved from below baseline on average in the early era to above baseline by the modern era, and its dispersion across coaches grew by about a third (standard deviation up 34%), suggesting experimentation as the tactic diffused. Whether the residual erosion reflects defensive adaptation (counter-strategies against predictable offensive aggression) or competitive equilibration (offensive aggression becoming table stakes) remains ambiguous.

5.2 Causality: Evidence from Within-Coach Variation

Our two-way fixed effects analysis provides evidence that the offensive aggression-performance relationship is at least partly causal. By comparing each coach to themselves across years, we find that a one standard deviation increase in offensive aggression corresponds to 0.44 additional WAR games (within-coach two-way fixed effects $\beta = 1.40$, 95% CI $[0.76, 1.96]$, $p < 0.001$; Cohen's $d = 0.23$), even after removing all between-coach differences and temporal trends. The relationship is not simply that better coaches happen to be more aggressive. Within the same coach, being more aggressive predicts better performance.

However, era-stratified within-coach estimates do not follow a monotone trend, preventing us from confirming whether this within-coach effect has eroded. Three findings converge on a causal interpretation. Cross-sectional correlations are strongest when tactics were rare, the relationship erodes as tactics spread, and a within-coach effect persists after controlling for stable characteristics. For practitioners, the practical reading is that offensive aggression's edge has narrowed to at most a weak association in the modern era.

5.3 Scheme Identity vs. Situational Decision-Making

Our analyses reveal a distinction between two types of coaching DNA, though not as clean as a simple binary. *Schematic identity*, meaning formation preferences and defensive packages, transfers through offensive coaching trees (shotgun OC: $r = 0.399$, 95% CI $[0.21, 0.54]$, $p < 0.001$) and persists through promotion (shotgun: $r = 0.674$, $[0.48, 0.80]$, $p < 0.001$; defensive aggression: $r = 0.570$, $[0.22, 0.82]$, suggestive at

$n = 17$). *Situational decision-making*, the in-game risk choices, predicts performance (offensive aggression: $r = 0.30$ at the career level, $[0.16, 0.45]$, $p < 0.001$) but is not transmitted through coaching trees ($r = 0.080$, $[-0.05, 0.22]$, $p = 0.24$).

Shotgun preference and defensive aggression bridge both categories. They transfer through coaching relationships *and* predict performance (shotgun: $r = 0.23$, 95% CI $[0.04, 0.41]$; defensive aggression: $r = 0.19$, $[0.02, 0.35]$, career level). This suggests that some schematic choices are not purely stylistic, since they carry competitive implications. Transmission also appears domain-specific, with only offensive coordinators transmitting offensive schematic identity (shotgun DC: $r = 0.096$, $[-0.12, 0.30]$, near zero), consistent with coaching DNA propagating through expertise rather than mere exposure.

The distinction also surfaces within offensive aggression. Fourth-down philosophy is among the most persistent offensive aggression components within coaches (Table 17), arguably the most "schematic" aspect of offensive aggression given that teams develop explicit decision charts, even though its cross-coach *inheritance* is weak once mentor clustering is applied (among components, only the pass-heavy OC inheritance remains). Pass-heavy and two-point tendencies are more reactive to game flow.

5.4 Challenging the Coaching Tree Narrative

The coaching tree narrative shapes NFL hiring decisions, but our data sit uneasily with its premises. Mentor WAR shows at most a weak association with protégé WAR ($r = 0.124$, 95% CI $[-0.01, 0.26]$, $p = 0.074$), giving little indication that working under a successful coach makes a coach successful. Composite offensive aggression shows little transmission ($r = 0.080$, $[-0.05, 0.22]$). For situational decisions, the premises of pedigree-based hiring find little support here.

However, offensive coaching trees *do* transmit schematic identity. Shotgun preference shows clear OC mentor-protégé inheritance ($r = 0.399$, $p < 0.001$), and coordinator-to-HC analysis indicates that formation preferences ($r = 0.674$) and defensive aggression ($r = 0.570$) persist through promotion, while pace does not reliably once genes are cross-fit. When Matt Nagy left Kansas City's shotgun-heavy offense to coach Chicago, he installed a nearly identical formation philosophy. When Dan Quinn left Dallas's aggressive defense to coach Washington, his defense looked similar. Teams can reliably predict the *system* a coordinator will install, but not whether they will win.

5.5 Limitations

Several limitations warrant discussion beyond the causality concerns addressed earlier.

First, our play-by-play data begin in 2006, missing earlier eras. We cannot determine whether the early aggressive advantage we observe in 2006-2011 was itself a continuation of an even longer trend or represented a genuine inflection point. Historical data would reveal whether similar adoption curves occurred for previous innovations.

Second, our inheritance analyses are limited by sample size. The career-average offensive aggression analysis uses 513 mentor-protégé pairs, while the direct coordinator-to-HC gene analysis uses 17–40 transitions depending on gene type. The defensive aggression analysis is particularly constrained (n=17) because defensive tracking data only became available in 2016. We address the small-cluster regime by reporting wild cluster bootstrap p-values for subgroups with fewer than 40 clusters. Even so, the shotgun coordinator-to-HC result (n=40, $r = 0.674$, $p < 0.001$) is robust, and the OC mentor-protégé analysis (n=249, $r = 0.399$, $p < 0.001$) corroborates the pattern, whereas the defensive-aggression transition (n=17) should be read as suggestive.

Third, the gene-WAR correlations should be interpreted with caution regarding causality. The defensive aggression-WAR correlation ($r = 0.188$, 95% CI $[0.02, 0.35]$, career level) may reflect reverse causation: teams with better defensive talent can afford to play man coverage and send extra rushers, whereas the gene framework attributes this to coaching philosophy. Our residual approach controls for game situation but cannot fully separate this preference from roster-enabled opportunity.

Fourth, we cannot observe coaching constraints. Some coaches may want to be aggressive but face organizational pressure to be conservative. Others may have coordinator partnerships that override their personal preferences. Our measures capture observed behavior rather than underlying intentions, which may differ due to unmeasured constraints.

Fifth, single-season WAR is itself a noisy outcome: its year-over-year reliability is only about one-quarter, so measurement error attenuates our season-level gene-WAR correlations. We mitigate this by anchoring the performance claim on career-level relationships (where mean WAR is far more reliable) and by inverse-variance weighting, but the season-level estimates should be read as conservative lower bounds rather than precise effect sizes.

Sixth, we attribute each gene to the head coach as the accountable decision-maker, yet play-calling responsibility varies: some head coaches call their own offensive or defensive plays while most delegate to a coordinator. This introduces idiosyncratic noise into the attribution rather than systematic bias, and the strong face validity and cross-team portability of the genes indicate that head-coach attribution nonetheless captures genuine coach-specific signal. Relatedly, the defensive-aggression gene is measured at the team level and so blends head coach, coordinator, and personnel.

Finally, "offensive aggression" is best understood as a formative index of four weakly-correlated facets (fourth-down, pass-run balance, deep passing, two-point) rather than a single latent trait; we therefore report the components alongside the composite, and note that the composite-WAR relationship is carried mainly by the pass-heavy and fourth-down facets.

6 Conclusion

Through analysis of more than 600,000 plays, 608 coach-years, and 612 mentor-protégé pairs (2006–2024), we develop a machine learning framework for measuring coaching "DNA" across four strategic dimensions and test what this DNA is, what it predicts, and how it transmits.

First, coaching genes are genuine, stable traits of the coach rather than artifacts of the team. Coach identity explains roughly five times as much of each gene's variance as team identity, and when coaches change teams their strategic tendencies travel with them (composite offensive aggression $r = 0.61$, 95% CI $[0.33, 0.78]$, across team changes), with rankings that match public reputation. This grounds the rest of the analysis: the genes measure the coach.

Second, most genes predict performance, though differently. Composite offensive aggression ($r = 0.189$, $p < 0.001$), shotgun formation ($r = 0.142$, $p = 0.004$), and defensive aggression ($r = 0.151$, $p = 0.015$) all correlate with WAR under coach-clustered inference, while tempo does not. Because single-season WAR is mostly noise, the more reliable anchor is the career level, where the offensive aggression relationship strengthens to $r = 0.30$. The offensive aggression edge has diminished over time, from $r = 0.24$ (2006–2011) to $r = 0.10$ (2018–2024), with a modest 2012 structural break ($p = 0.03$), as analytics-driven tactics spread.

Third, coaching trees transmit systems more than success or risk tolerance. Mentor WAR shows at most a weak association with protégé WAR ($r = 0.124$, $p = 0.074$), and composite offensive aggression shows little inheritance ($r = 0.080$). But shotgun formation preference transmits through offensive coaching trees ($r = 0.399$, $p < 0.001$), though not defensive ones ($r = 0.096$), and direct coordinator-to-HC analysis indicates that schematic identity persists through promotion, with shotgun ($r = 0.674$) and defensive aggression ($r = 0.570$) transmitting strongly while pace and situational aggression do not. Organizations can far more readily predict the system a coordinator will install than whether they will win.

Within-coach fixed-effects analysis adds causal support for the performance result (0.44 WAR games per standard deviation of offensive aggression; within-coach $\beta = 1.40$, 95% CI $[0.76, 1.96]$, $p < 0.001$), even as the cross-sectional edge has narrowed in the modern era. The frontier appears to have moved beyond offensive aggression frequency, perhaps to scheme innovation, within-game adaptation speed, or player development, dimensions we are only beginning to measure.

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